

Kian Shams

01-09-2001

Dutch

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EDUCATION

- **University of Amsterdam** Amsterdam, The Netherlands
Theoretical physics (MSc, cum laude) *September 2024 - Present*
 - **Thesis title:** [Toric Calabi–Yau threefolds & Topological Strings](#)
 - **Supervisor:** Dr M. L. Vonk
 - **Thesis grade:** 9.0/10
 - **GPA:** 8.1/10
 - **Extracurricular:** Algebra 2 (ring theory)
- **Leiden University** Leiden, The Netherlands
Theoretical physics (MSc) *September 2023 - July 2024*
 - Transferred to Amsterdam to pursue a curriculum with a stronger emphasis on the mathematical foundations of theoretical physics.
 - **Completed courses:** Quantum Field Theory, Effective Field Theory, Quantum Information, Quantum Theory, Origin & Structure of the Standard Model, Topics in Theoretical Physics (cosmology), Computational Physics (and Academic & Professional Skills).
 - **Intermediate GPA:** 7.9/10
 - **Extracurricular:** Introduction to Measure Theory, Projective Geometry
- **Technical University Delft** Delft, The Netherlands
Applied physics (BSc) *September 2019 - December 2022*
 - **Thesis title:** Gate Set Tomography on a qutrit in a Nitrogen-Vacancy centre
 - **Supervisor:** Dr J. Borregaard
 - **Thesis grade:** 8.5/10
 - **GPA:** 7.6/10

SCHOOLS & WORKSHOPS

- **ICTP** Trieste, Italy
Winter school on number theory and physics *November 2025*
 - **Topics:** Hodge theory, Arithmetic geometry of periods, Harmonic Maaß forms, K3 surfaces, Automorphic forms, Hypergeometric functions and local systems

WORK EXPERIENCE

- **UvA** Amsterdam, The Netherlands
TA for Electrodynamics & student mentor *October 2025 - June 2026*
 - **Responsibilities:** Assisting students in the problem sessions, making part of the exam and grading. Being a student mentor entails guiding first-year master's students through the programme and providing support with academic and practical matters.
- **Stichting Studiebegeleiding Leiden (SSL)** Leiden, The Netherlands
Teacher mathematics *March 2023 - July 2026*
 - **Responsibilities:** Helping students in their final year of secondary school prepare for their mathematics final exam.
- **TU Delft** Delft, The Netherlands
TA positions *October 2020 - June 2022*
 - **Courses:** Calculus 2, Electricity & Magnetism, Quantum Mechanics III
 - **Responsibilities:** Assisting students in the problem sessions with homework problems and explaining course content.

SKILLS

- Intermediate proficiency in Python (plotting, numerical integration, differential equations) and basic proficiency in Mathematica (symbolic manipulation and integration).